

Sophia Kaltsouni Mehdizadeh

Ph.D. Candidate

University of Colorado Boulder

I am a music cognition researcher with a strong technical background and extensive experience designing and conducting EEG, fMRI, and behavioral research studies. I have worked with quantitative, qualitative, and mixed methods, as well as using participant-selected and individualized stimuli.

I. Education

April 2027	University of Colorado Boulder <i>Ph.D. in Creative Technology & Design, Cognitive Science, Neuroscience</i> Advisor: Dr. Grace Leslie Neuroscience co-supervisor: Dr. Marie Banich
December 2021	Georgia Institute of Technology <i>M.S. in Music Technology</i> Advisors: Dr. Grace Leslie, Dr. Thackery I. Brown
April 2018	University of Michigan <i>B.S.E. in Electrical Engineering, Music Minor</i>

II. Employment History

Present – Jan 2023	University of Colorado Boulder, ATLAS Institute Brain Music Lab <i>Pre-doctoral Researcher</i> Supervisor: Dr. Grace Leslie
Sept 2022 – June 2022	Microsoft Research, Brain Computer Interfaces Group <i>PhD Research Intern</i> Supervisors: Dr. Yu Te Wang, Dr. Ivan Tashev
June 2022 – Jan 2020	Georgia Tech, School of Music, Brain Music Lab <i>Graduate Research Assistant</i> Supervisor: Dr. Grace Leslie
Jan 2020 – June 2018	Mitsubishi Electric Automotive America <i>Advanced Development Audio Engineer</i>

III. Honors and Awards

1. University of Colorado Boulder Graduate School Dissertation Completion Fellowship, 2026 (\$15,711).
2. Philanthropic Educational Organization (PEO) Scholar Awards local chapter nominee, 2024.
3. Student research award, Society for Music Perception and Cognition, 2024 (\$290).
4. Moog Music Hackathon winner, 2021 (\$3,000).

IV. Research, Scholarship, and Creative Activities

A. Refereed Publications and Submitted Articles

All publications below are in peer-reviewed journals or conference proceedings. Authors are ordered by contribution level, with the primary contributors listed first. Laboratory directors are listed last by convention.

(*equal contribution)

A1. Published and Accepted Journal Articles

1. Ren, Y.*, **Mehdizadeh, S. K.***, Leslie, G., & Brown, T. I. (2024). Affective Music During Episodic Memory Recollection Modulates Subsequent False Emotional Memory Traces: An fMRI Study. *Cognitive, Affective, & Behavioral Neuroscience*, 24(5), 912-930. <https://doi.org/10.3758/s13415-024-01200-0>
2. **Mehdizadeh, S. K.**, & Leslie, G. (2023). The Physiology of Musical Preference: A Secondary Analysis of the Study Forrest Dataset. *Music Perception*, 40(5), 395-409. <https://doi.org/10.1525/mp.2023.40.5.395>

A2. Conference Presentation with Proceedings (Refereed)

1. **Mehdizadeh, S. K.**, Cutrell, E., Winters, R. M., Djuric, N., Cheng, Y., Tashev, I. J., & Wang, Y. T. (2023). EEG and Eye-Tracking Error-Related Responses During Predictive Text Interactions: A BCI Case Study. *45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC '23)*. <https://doi.org/10.1109/EMBC40787.2023.10340598>

A3. Other Refereed Material

1. (Poster) **Mehdizadeh, S. K.**, Ren, Y., Brown, T. I., & Leslie, G. (2022). [Using Music to Modulate Human Emotional Memory](#). *2022 Biennial Meeting of the Society for Music Perception and Cognition (SMPC '22)*.

A4. Journal Articles in Preparation

1. **Mehdizadeh, S. K., & Leslie, G. (2026).** *"Finding this song was part of becoming who I am now:" A Mixed-Methods Investigation of Music Appreciation in Emerging Adults.*
2. **Mehdizadeh, S. K., & Leslie, G. (2026).** Neural Mechanisms of Self-Referential and Sensory-Motor Music Appreciation in Emerging Adults.

B. Other Publications and Creative Products

B1. Other Academic Works

1. **Mehdizadeh, S. K. (2021).** Using Music to Modulate Emotional Memory. [Master's thesis, Georgia Institute of Technology].
<http://hdl.handle.net/1853/66164>

B2. Creative Works

1. [*Idiophone: A Resonant Sound Installation*](#), as part of MediaLive: Technology as Healing, **Boulder Museum of Contemporary Art** (premiered October 12th, 2023 in Boulder, Colorado).
2. [*Global Pandemic: An EEG Biofeedback Performance Piece*](#), **Georgia Tech School of Music** (premiered May 4th, 2022 in Atlanta, Georgia).

C. Presentations

1. **Mehdizadeh, S. K., & Leslie, G. (2026).** [*Forms of music appreciation, social context, and recommendation source uniquely impact felt experience.*](#) 2026 Biennial Meeting of the Society for Music Perception and Cognition (SMPC '26).
2. **Mehdizadeh, S. K., Ren, Y., Leslie, G., Brown, T. I. (2024).** [*Emotional Music Modulates Episodic Memory During Recollection and Induces False Traces.*](#) 2024 Biennial Meeting of the Society for Music Perception and Cognition (SMPC '24).
3. Roque, T. R., Rajagopalan, N., Jain, S., **Mehdizadeh, S. K., & Leslie, G. (2022).** Multimodal, Musical Hyperscanning to Promote Empathy in HCI. *Empathy-Centric Design At Scale Workshop at CHI 2022.*
4. **Mehdizadeh, S. K., & Leslie, G. (2021).** [*Novel Methodologies for Secondary Analyses of Physiological and Musical Preference Data.*](#) *International Conference on Music Perception and Cognition (ICMPC 16).*

D. Funding

D1. As Principal Investigator

1. University of Colorado Boulder graduate student travel grant (2026). Title: Forms of music appreciation, social context, and recommendation source uniquely impact felt experience. Amount: \$650.
2. University of Colorado Boulder Institute of Cognitive Science student research grant (2026). Title: *Neural Mechanisms of Music Appreciation*. Amount: \$1,500.
3. University of Colorado Boulder Beverly Sears graduate student research grant (2025). Title: *Mechanisms of Music Appreciation and Shared Listening*. Amount: \$850.
4. University of Colorado Boulder Center to Advance Research and Teaching in the Social Sciences graduate student seed grant (2024). Title: *Mechanisms of Music Appreciation and Shared Listening*. Amount: \$1,200.
5. University of Colorado Boulder Institute of Cognitive Science student research grant (2024). Title: *Mechanisms of Music Appreciation and Shared Listening*. Amount: \$800.
6. University of Colorado Boulder Institute of Cognitive Science student travel grant (2024). Title: *Emotional Music Modulates Episodic Memory During Recollection and Induces False Traces*. Amount: \$1,000.
7. University of Colorado Boulder ATLAS Institute travel award (2024). Title: *Emotional Music Modulates Episodic Memory During Recollection and Induces False Traces*. Amount: \$650.
8. Boulder Museum of Contemporary Art artist honorarium (2023). Title: *Idiophone: A Resonant Sound Installation*. Amount: \$800.

D2. As Trainee

1. GSU/Georgia Tech Center for Advanced Brain Imaging Seed Grant (2021). Title: *Using Music to Modulate Emotional Memory*. Amount: 20 hours of MRI scanning.

E. Media

1. Boulder Daily Camera (2023): [Future's so bright: Revolutionary new media art shines at BMoCA's 'MediaLive: Technology as Healing' exhibit](#)

F. Other Professional Activities

F1. Workshop Acceptances

1. **NextProf Nexus 2025**, *UC Berkeley* (participation fully funded).
 - *Topic*: Navigating the academic job market and succeeding as early career faculty.
2. **NeuroHackademy 2024**, *University of Washington eScience Institute*.
 - *Topic*: Best practices for analysis and reproducibility in human neuroscience.

V. Teaching and Mentorship

A. Graduate Teaching Assistantships

1. Wearable Technologies, ATLAS Institute, University of Colorado Boulder (Spring 2025)
 - *Topic*: Design and prototyping of new wearables using sensors and microcontrollers.
 - *Supervisor*: Dr. Shaz Zamore
2. Technology Ensemble, Music Technology, School of Music, Georgia Tech (Fall 2022)
 - *Topic*: Designing and building electronic musical interfaces and systems for performance.
 - *Supervisor*: Dr. Jeremy Muller
3. Project Studio Tech, Music Technology, School of Music, Georgia Tech (Fall 2022)
 - *Topic*: Quantitative data analysis methods using Python.
 - *Supervisor*: Dr. Peter Knees
4. Intro to Music Theory, Music Technology, School of Music, Georgia Tech (Summer 2020)
 - *Topic*: Basic notation, scales, roman numeral analysis, and voice leading.
 - *Supervisor*: Dr. Frank Clark

B. Individual Student Guidance

B1. Master's Students

1. Naimisha Sanjay, *Brain Music Lab student researcher* (Ongoing). Role: research mentor.
2. Wesley Hughes, *Brain Music Lab student researcher* (Ongoing). Role: research mentor.

B2. Undergraduate Students

1. Marissa Khan, *Intermountain Neuroimaging Consortium research intern* (2026).
Role: MRI research study mentor.
2. Estrella Ordaz, *Intermountain Neuroimaging Consortium research intern* (2026).
Role: MRI research study mentor.
3. Sophia Montie, *Creative Technology & Design independent study project* (2025).
Now a master's student at NC State. Role: co-supervisor.

B3. High School Students

1. Sophia Pych, *International Science & Engineering Fair project* (2024). Now an undergraduate student at Georgia Tech. Role: professional research supervisor.

VI. Service

A. Institute Contributions

May 2026 – Jan 2024	University of Colorado Boulder, Institute of Cognitive Science <i>Student Grants Committee, ATLAS Institute representative</i>
Dec 2022 – Aug 2021	Women in Music Technology @ Georgia Tech <i>Treasurer and Chartering Leader</i>